

LITERATURE SURVEY

Project Title

Machine Learning-Based Loan Risk Prediction: A Comparative Study of Ensemble, Oversampling, and Topological Approaches for Credit and Loan Default Assessment

Prepared By

RITHISH R(23CSR178)

NAVEEN SRINIVAS V(23CSR145)

MENAHA R(23CSR130)

**Department of Computer Science and Engineering
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INTRODUCTION

Access to credit plays an important role in supporting individuals, businesses, and economic growth. Before approving a loan, financial institutions must evaluate whether a borrower is capable of repaying it. Traditional credit risk assessment methods, such as logistic regression and rule-based scorecards, are widely used because they are simple and easy to interpret. However, these methods often struggle to handle large datasets, complex relationships between borrower attributes, and imbalanced data, which can reduce prediction accuracy.

Recent advancements in Artificial Intelligence and Machine Learning have significantly improved loan risk prediction. Algorithms such as Decision Tree, Random Forest, XGBoost, LightGBM, and Neural Networks can identify hidden patterns in financial data and provide more accurate predictions. In addition, techniques like data balancing, feature engineering, and explainable AI help improve model performance and reliability. This literature survey reviews five recent research papers on loan and credit risk prediction, compares the methods used, highlights their advantages and limitations, and provides the foundation for developing our proposed machine learning-based loan risk prediction system.

OBJECTIVES OF LITERATURE SURVEY

- To study recent research on machine learning techniques for loan and credit risk prediction.
- To understand various algorithms and feature engineering strategies used for assessing default risk.
- To examine methods for handling imbalanced credit and loan default data.
- To analyze the advantages and limitations of existing loan risk prediction models.
- To compare the performance of different machine learning algorithms across diverse lending domains (personal credit, SME credit, and maritime finance).
- To understand the role of explainability techniques (e.g., SHAP) in improving model transparency for credit decisions.
- To identify existing research gaps and challenges in loan risk prediction.
- To provide the background required for the proposed loan risk prediction project.

DETAILED LITERATURE SURVEY (PAPER 1–PAPER 5)

Paper 1: Design of a Personal Credit Risk Prediction Model and Legal Prevention of Financial Risks (Zhuang & Wei, 2024)

This paper presents a machine learning-based approach for improving personal credit risk prediction. The authors addressed the problem of imbalanced credit data by using a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) to generate additional default loan samples and combined it with a Light Gradient Boosting Machine (LightGBM) model for classification. The proposed model achieved better performance than Extreme Gradient Boosting (XGBoost) and Gradient Boosting Decision Tree (GBDT) models, with an Area Under the Receiver Operating Characteristic Curve (AUC) of 0.86 and a Kolmogorov–Smirnov (KS) Statistic of 0.87. The study also identified important features such as the Annuity-to-Credit Ratio, External Source Score 3 (EXT_SOURCE_3), and Days_Birth for predicting loan defaults. The authors concluded that the model provides accurate credit risk prediction, but future work should focus on larger datasets and improving model interpretability.

Paper 2: Investigation of the Application of Machine Learning Algorithms in Credit Risk Assessment of Medium and Micro Enterprises (Zhao, 2024)

This paper focuses on improving credit risk assessment for Medium and Micro Enterprises (MMEs), which often have limited and unstable financial records. The author proposed a Consistent Assessment Method (CAM) based on Linear Learning Analysis (LLA) to identify risky financial patterns by analyzing changes in credit and financial growth over different reporting periods. The model was tested using financial data from food and beverage manufacturing enterprises and compared with Firefly Algorithm–Support Vector Machine (FA-SVM), Weighted Synthetic Minority Oversampling Technique with Expectation Maximization (WSMOTE-EM), and Support Vector Machine–Backpropagation (SVM-BP) models. The results showed that CAM-LLA achieved higher risk detection accuracy, improved financial revision accuracy, and reduced classification time. The authors concluded that the model is simple, interpretable, and effective for credit risk assessment, but its linear nature makes it less suitable for handling complex non-linear financial data.

Paper 3: A Novel Hybrid Model for Loan Default Prediction in Maritime Finance Based on Topological Data Analysis and Machine Learning (Kheneifar & Amiri, 2025)

This paper proposes a hybrid approach for predicting loan defaults in the maritime finance sector, where traditional credit risk models often fail to capture complex financial relationships. The authors combined Topological Data Analysis (TDA) with Machine Learning (ML) by extracting network-based features and using them to train models such as Graph Neural Network (GNN), Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), Gradient Boosting (GB), Adaptive Boosting (AdaBoost), and Extreme Gradient Boosting (XGBoost). The results showed that adding topological features significantly improved prediction accuracy, with XGBoost achieving a Receiver Operating Characteristic – Area Under the Curve (ROC-AUC) score of 0.973. The study also used SHapley Additive exPlanations (SHAP) to identify important factors affecting loan defaults, including loan amount, vessel age, debt-to-equity ratio, and interest rate spread. The authors concluded that the proposed model improves loan default prediction by capturing complex risk patterns, but future work should include larger datasets and additional non-financial factors such as economic and geopolitical conditions.

Paper 4: A Novel Method for Credit Scoring Based on Cost-Sensitive Neural Network Ensemble (Yotsawat, Wattuya & Srivihok, 2021)

This paper proposes a Cost-Sensitive Neural Network Ensemble (CS-NNE) to improve credit scoring on imbalanced datasets without using data resampling techniques. The model assigns different misclassification costs to multiple neural networks and combines their predictions using majority voting. An Autoencoder (AE) is also used to detect and remove noisy data before training. The proposed model was evaluated on five real-world credit datasets and compared with Random Forest (RF), Extreme Gradient Boosting (XGBoost), Bagging, and Adaptive Boosting (AdaBoost). The results showed that CS-NNE achieved better Area Under the Receiver Operating Characteristic Curve (AUC), higher default detection rate, and lower misclassification cost. The authors concluded that the model is effective in identifying risky borrowers, although it requires further testing on different datasets and imbalance conditions.

Paper 5: Machine Learning and Deep Learning for Loan Prediction in Banking: Exploring Ensemble Methods and Data Balancing (Sayed, Alabrah, Rahouma, Zohaib & Badry, 2024)

This paper compares different Machine Learning (ML) and Deep Learning (DL) models for bank loan prediction using a large loan dataset. The authors tested algorithms such as Random Forest (RF), Decision Tree (DT), Logistic Regression (LR), K-Nearest Neighbors (KNN), and deep learning models including Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), Residual Network (ResNet), and Dense Convolutional Network (DenseNet). By applying Synthetic Minority Oversampling Technique (SMOTE) and SMOTE-TOMEK for data balancing, DenseNet achieved the best performance with 92% accuracy and 99% Area Under the Receiver Operating Characteristic Curve (AUC-ROC). The study concluded that data balancing techniques significantly improve loan prediction accuracy, although advanced deep learning models require more computational resources.

COMPARATIVE SUMMARY OF REVIEWED PAPERS

Paper	Domain	Method	Key Result	Main Limitation
Paper 1 (Zhuang & Wei, 2024)	Personal consumer credit	WGAN-GP oversampling + LightGBM	86.7% avg. accuracy; AUC 0.86, KS 0.87	Limited dataset; needs broader validation
Paper 2 (Zhao, 2024)	Medium/micro enterprise credit	CAM using Linear Learning Analysis (LLA)	14.65% higher risk detection; 5.56% higher financial revisions	Linear model struggles with non-linear/high-frequency data
Paper 3 (Kheneifar & Amiri, 2025)	Maritime shipping loans	Topological Data Analysis + GNN + ML classifiers	ROC-AUC up to 0.973 (XGBoost); LR AUC 0.64→0.92	Region-specific data; excludes geopolitical variables
Paper 4 (Yotsawat et al., 2021)	General credit scoring (bank + public datasets)	Cost-Sensitive Neural Network Ensemble (CS-NNE)	+1.36% AUC, +15.67% DDR, +6.11% GM over single NN; MC 0.109	Not always highest accuracy/specificity; needs testing on more base algorithms
Paper 5 (Sayed et al., 2024)	Bank loan approval prediction	5 ML + 8 DL models with Ensemble Methods and SMOTE/SMOTE-TOMEK	DenseNet/ResNet best: 92% accuracy, 99% AUC-ROC, MCC 0.64	Single Kaggle dataset; deep models need heavy compute

SUMMARY

The reviewed research papers show that Machine Learning (ML) and Deep Learning (DL) techniques can significantly improve loan risk prediction compared to traditional methods. Different approaches such as Light Gradient Boosting Machine (LightGBM), Extreme Gradient Boosting (XGBoost), Random Forest (RF), Graph Neural Network (GNN), Residual Network (ResNet), and Dense Convolutional Network (DenseNet) achieved high prediction accuracy. The studies also highlight the importance of handling imbalanced datasets using techniques like Synthetic Minority Oversampling Technique (SMOTE), SMOTE-TOMEK, Generative Adversarial Networks (GANs), and cost-sensitive learning to improve the detection of loan defaulters.

Overall, these papers indicate that combining effective data balancing methods, powerful ensemble learning algorithms, and explainable Artificial Intelligence techniques leads to better loan risk prediction. Based on these findings, our proposed project aims to develop a reliable loan risk prediction model that provides accurate predictions, improves loan approval decisions, and helps financial institutions reduce the risk of loan defaults.